

Pose and Wheeled-Robot Motion

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1. Overview

Key point: A robot can only set its wheel speeds, but every task is stated in places: “go to the door”. A model of the robot’s motion connects the two, so that we can predict where wheel commands take the robot and choose the commands that reach a goal.

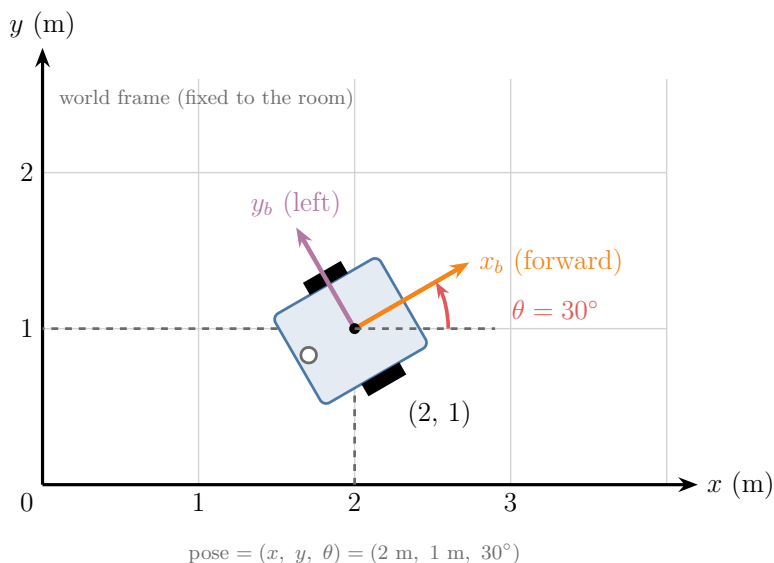


Figure 1: A small two-wheeled robot on a floor grid at $x = 2 \text{ m}$, $y = 1 \text{ m}$, facing 30 degrees left of the x-axis, with its own forward and left axes drawn on it

The only thing a wheeled robot can do is spin its motors. Everything we want from it is about places: drive to the door, stay in the lane, park in the corner. So we need a model that answers two questions, one in each direction:

1. **Prediction:** if the wheels spin at these speeds, where will the robot be in two seconds?
2. **Control:** to get to the door, at what speeds should the wheels spin?

Both questions need a precise way to say where the robot is, and a rule linking wheel speeds to changes in that position. In this Note we build both for the robot in Figure 1, which has two driven wheels like most indoor robots:

- three numbers that say where the robot is, and why it takes exactly these three (Section 2);
- why a rolling wheel’s spin tells us its speed over the floor (Section 3);
- the link from the two wheel speeds to the robot’s forward speed and turn rate, derived from one fact about turning bodies, and the link back (Section 4);
- how those speeds move the robot over the floor, and how to predict its path step by step (Section 5);
- why the robot cannot slide sideways, and why it can still reach every pose (Section 6);
- why cars follow the same model with stricter limits (Section 7);

- why a trailer swings into line when pulled forward but folds up in reverse (Section 8).

2. Pose: where the robot is

Key point: Position alone does not tell us where the robot will go next; we also need the direction it faces. Position (x, y) plus heading θ , measured in an agreed frame, is the pose.

2.1 Why we need a frame

“The robot is at $(2, 1)$ ” means nothing until we say measured from where, and along which directions. Two people measuring from different corners of the room would give the same robot different numbers. So we first agree on a reference: a corner of the room as the origin, with an x-axis along one wall and a y-axis along the other. Such a choice of origin and axes is a **coordinate frame** (G-2281). The frame fixed to the room is the **world frame** (G-2282); every position in this Note is measured in it.

2.2 Why position is not enough: the heading

Take two robots at the same spot, $(2, 1)$, both driving forward at 0.5 m/s for one second. One faces along the x-axis, the other along the y-axis:

Robot faces	Where it is after 1 s
along x	$(2.5, 1)$
along y	$(2, 1.5)$

Same position, same speed, different result. To predict motion we must also record the direction the robot faces, its **heading** (G-2284): the angle θ from the world x-axis to the robot’s forward direction.

Angles are measured counter-clockwise, so a robot facing along y has $\theta = 90^\circ$. That direction is a convention, not a law of nature. The robotics software standard ROS fixes it (x forward, y left, positive angles counter-clockwise when seen from above) so that every program reads the numbers the same way (REP-103).

Together, position and heading are the robot’s **pose** (G-2280). For the robot in Figure 1:

$$x = 2 \text{ m}$$

$$y = 1 \text{ m}$$

$$\theta = 30^\circ$$

2.3 Why angles are measured in radians

Key point: In radians, the arc a turning point travels is simply radius times angle, with no conversion factor; every formula in this Note relies on that.

We will often need this: a point at distance d from a centre turns by some angle; how far does it travel along its arc? A full turn travels the whole circumference, $2\pi d$, and a part of a turn travels the same part of the circumference.

Try it in degrees, with $d = 0.5$ m and a turn of 30° . Thirty degrees is one twelfth of a full turn:

$$\begin{aligned}\frac{30}{360} \times 2\pi \times 0.5 \\ = 0.262 \text{ m}\end{aligned}$$

Multiplying the radius by the angle in degrees gives nonsense:

$$0.5 \times 30 = 15 \text{ m}$$

Degrees need the extra factor $2\pi/360$ every time. So we measure angles in a unit that already contains it: the **radian** (G-2285), where a full turn is 2π radians. Then 30° is:

$$30 \times \frac{2\pi}{360} = 0.524 \text{ rad}$$

and the arc is radius times angle, with nothing else:

$$\text{arc} = d \times \Delta\theta$$

$$0.5 \times 0.524 = 0.262 \text{ m}$$

One radian is the angle at which the arc is exactly as long as the radius. This arc rule is the reason the wheel formula (Section 3), the turning formulas (Section 4) and the cosine and sine functions of Python's `numpy` all work in radians.

2.4 Why the robot also has its own frame

A robot's sensors see the world from the robot's point of view: a distance sensor reports "an obstacle 1 m straight ahead", not a position on the room's map. To mark that obstacle on the map, we need to know where the robot is and which way it faces:

Robot pose	“1 m straight ahead” is at
(2, 1), facing along x	(3, 1)
(2, 1), facing along y	(2, 2)

So the robot carries a frame of its own, the **body frame** (G-2283): origin at the middle point between its two wheels, x-axis pointing forward, y-axis pointing to its left (Figure 1). It moves and turns with the robot. The pose says exactly where the body frame sits inside the world frame, and that is what lets us move any measurement between the two. The next Note in this chapter does those conversions in general.

2.5 Configuration and configuration space

Key point: The configuration is the full list of numbers that places the robot; all its possible values form a space, and later planning Notes search that space for paths.

A **configuration** (G-2286) is a list of numbers that pins down where every point of a robot is (MR Definition 2.1). Our robot is rigid, so its pose is enough: knowing where the middle of the axle is and which way the robot faces fixes every other point of the body. We write it as one vector q (a list of numbers):

$$q = (x, y, \theta)$$

$$q = (2, 1, 0.524)$$

The smallest number of values that does this is the robot’s **degrees of freedom** (G-2310): 3 for a robot on a floor. Two values would not be enough, as the table in Section 2.2 shows.

All possible configurations together form the **configuration space** (G-2287), or C-space (MR Definition 2.1). It gets a name because later Notes plan routes by searching it: every point of the C-space is one placement of the whole robot, and a route is a path through it. Two things about its shape matter:

- x and y can be any real numbers (any spot on an endless floor);
- θ wraps around, because turning by a full 2π brings the robot back to the same heading. So the headings 350° and 10° are only 20° apart, not 340° . A program that subtracts headings without allowing for the wrap-around turns the robot the long way round.

3. The differential-drive robot

Key point: Two motorised wheels on one axle are enough to drive anywhere and even to spin on the spot. If a wheel rolls without slipping, its speed over the floor is its radius times its spin rate.

3.1 Why two wheels and a caster

A **differential-drive** (G-2288) robot has two driven wheels on a common axle, each turned by its own motor. It is the most common way to drive an indoor robot (LaValle §13.1.2.2), because two motors are enough for everything: driving, turning and spinning on the spot (Section 4.4). Two wheels on one axle would tip over, so a third, free-swivelling **caster wheel** (G-2289), like the wheel under an office chair, holds the robot up without steering it (LaValle §13.1.2.2).

There is no steering wheel. The robot turns only through the **difference** between its two wheel speeds, and that is where the name “differential” comes from.

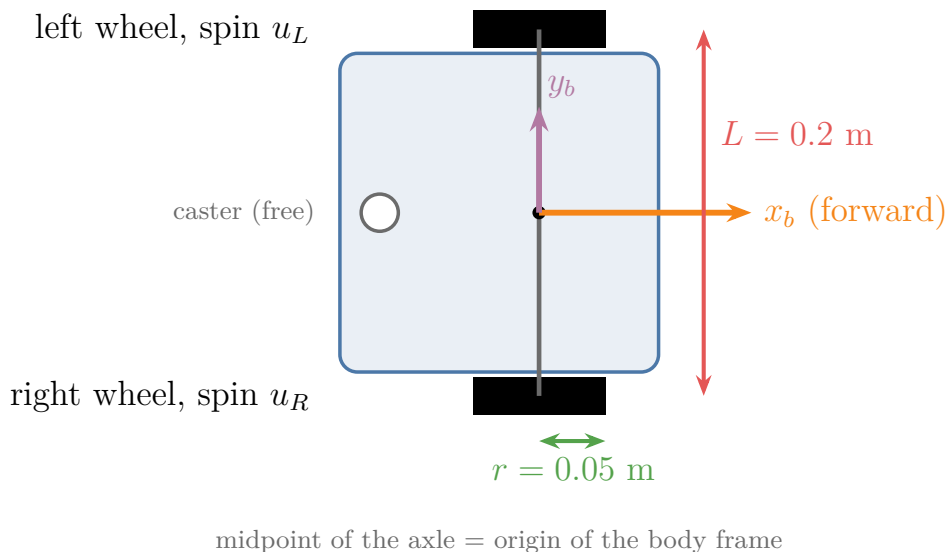


Figure 2: Top view of a differential-drive robot: right and left wheels of radius r , the distance L between them, the caster at the back, and the body frame at the midpoint of the axle

Figure 2 shows the only two measurements the model needs, with the values we use throughout. Both can be measured with a ruler:

$$r = 0.05 \text{ m} \quad (\text{wheel radius})$$

$$L = 0.2 \text{ m} \quad (\text{wheel separation})$$

3.2 Why wheel spin gives ground speed: rolling without slipping

Key point: A wheel that does not skid lays its rim down on the floor as it turns, so it moves forward by exactly the length of rim it turns.

A motor tells us how fast the wheel spins, in radians per second. What we need is how fast the robot moves over the floor, in metres per second. The link is the wheel's grip.

A wheel **rolls without slipping** (G-2290) when the point touching the floor does not skid. Each bit of rim that comes down touches the floor once and stays put while touching. So the wheel moves forward by exactly the length of rim it has turned (Figure 3). One full turn lays down the whole circumference:

$$2\pi \times r$$

$$= 6.283 \times 0.05$$

$$= 0.314 \text{ m}$$

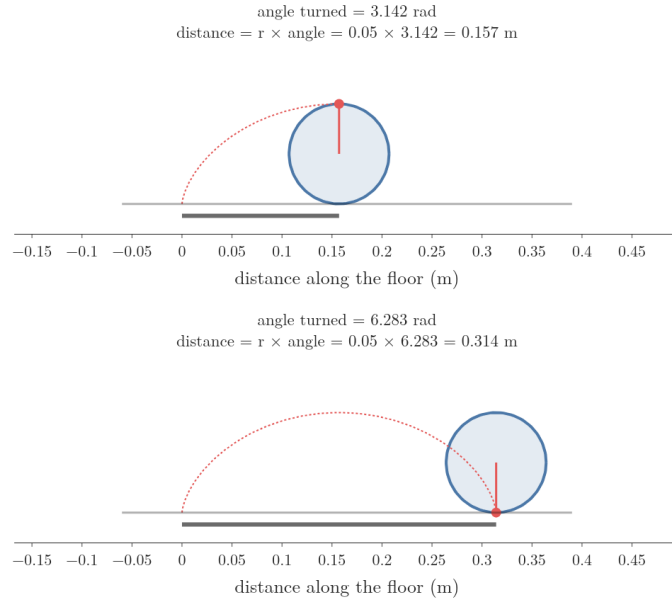


Figure 3: A wheel of radius 0.05 m rolls one full turn along the floor; a marked point on its rim traces the turn, and the wheel ends 0.314 m further on

A part of a turn lays down the same part of the rim, which is the arc rule of Section 2.3: radius times angle. So a wheel spinning at a rate u (radians per second) moves over the floor at:

$$v_{\text{wheel}} = r u$$

We write u_R and u_L for the spin rates of the right and left wheels, and v_R and v_L for their ground speeds. With the right wheel spinning at 12 rad/s and the left at 8 rad/s:

$$v_R = 0.05 \times 12 = 0.6 \text{ m/s}$$

$$v_L = 0.05 \times 8 = 0.4 \text{ m/s}$$

Extra: Real wheels slip a little, more on smooth floors and in fast turns. The formula then overestimates the distance moved, and a robot that tracks its position by counting wheel turns slowly drifts away from the truth. Later Notes on motion models and wheel odometry deal with that error.

4. From wheel speeds to robot speeds and back

Key point: The robot’s forward speed is the average of the two wheel speeds, and its turn rate is their difference divided by the wheel separation. Both follow from one fact: a turning rigid body circles a single point.

4.1 Why we describe motion by forward speed and turn rate

Wheel speeds are what the motors take, but they are an awkward way to think. Asked how to drive to the door, nobody says “right wheel 0.35 m/s, left wheel 0.25 m/s”; we say “not too fast, and bear left”. So we describe the motion of the whole robot with two numbers that match that way of thinking:

- its **forward speed** (G-2291) v : how fast the midpoint between the wheels moves along the heading, in metres per second;
- its **turn rate** (G-2292) ω (omega): how fast the heading θ changes, in radians per second, positive to the left.

A controller decides v and ω ; this section finds the formulas that convert them to wheel speeds and back.

4.2 Why the robot turns about a point on its axle line

Key point: Each wheel can only move straight ahead, square to the axle. The only centre the whole body can turn about that allows this lies on the axle line.

When a rigid body turns, every point of it moves on a circle around one centre, and its velocity at each moment points square to the line from that centre. Our two wheels must move straight ahead, because they cannot slide sideways (Section 6). Straight ahead is square to the axle. So the line from the centre to each wheel must be the axle line itself: the centre lies somewhere on the line through the axle (MR §13.3.1).

That centre is the turning point, also called the instantaneous centre of curvature. Its distance from the robot’s midpoint is the **turning radius** (G-2293) R .

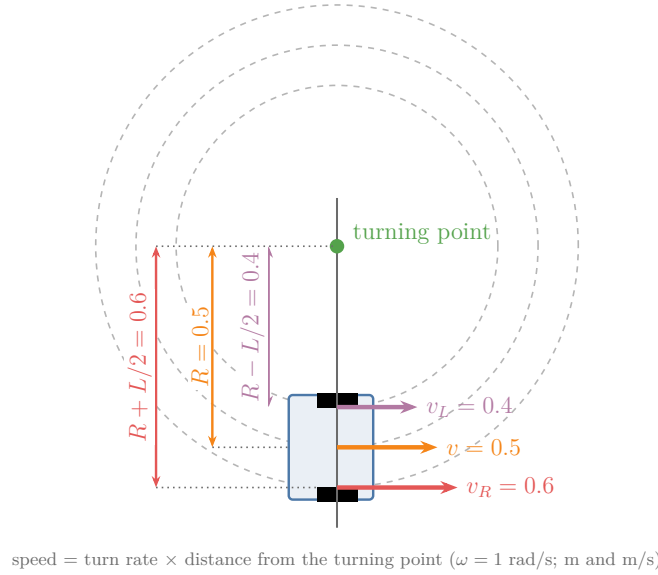


Figure 4: The robot turning about a point on its axle line. The right wheel is at distance $R + L/2$, the midpoint at R , the left wheel at $R - L/2$; their velocity arrows grow in proportion to the distance

The whole body turns by the same angle in each second, the turn rate ω . By the arc rule of Section 2.3, a point at distance d from the centre then travels ω times d metres per second. Points farther out move faster (Figure 4):

$$\text{speed} = \omega \times \text{distance}$$

For a left turn, the right wheel sits half the wheel separation farther out than the midpoint, and the left wheel half the separation closer in:

$$v_R = \omega (R + L/2)$$

$$v_L = \omega (R - L/2)$$

$$v = \omega R$$

4.3 Wheel speeds to forward speed and turn rate

Key point: Subtracting the two wheel equations isolates ω ; adding them isolates v .

Turn rate. Subtract the left-wheel line from the right-wheel line. The ωR parts cancel, leaving only the separation:

$$v_R - v_L = \omega (L/2 + L/2)$$

$$v_R - v_L = \omega L$$

$$\omega = \frac{v_R - v_L}{L}$$

The formula says why a wide robot turns more lazily: the same speed difference has to swing a longer axle. With a speed difference of 0.2 m/s:

Wheel separation L	Turn rate ω
0.2 m	1 rad/s
0.4 m	0.5 rad/s

Forward speed. Add the two lines. The $\omega L/2$ parts cancel:

$$v_R + v_L = 2\omega R$$

$$v_R + v_L = 2v$$

$$v = \frac{v_R + v_L}{2}$$

The midpoint moves at the average speed of the wheels because it sits halfway between them.

Worked example. With our wheel speeds, the forward speed is:

$$v = \frac{0.6 + 0.4}{2} = 0.5 \text{ m/s}$$

The turn rate is:

$$\omega = \frac{0.6 - 0.4}{0.2} = 1 \text{ rad/s}$$

The turning radius follows from $v = \omega R$:

$$R = v/\omega$$

$$R = 0.5/1 = 0.5 \text{ m}$$

Going from the wheel speeds to the robot's motion is the **forward kinematics** (G-2294) of the robot (LaValle §13.1.2.2). It answers half of the prediction question of Section 1; Section 5 finishes it.

4.4 Four ways to drive

Key point: Equal wheel speeds drive straight, opposite speeds spin on the spot, one stopped wheel pivots about that wheel, and anything else drives on a circle.

The two formulas predict four kinds of motion. Figure 5 plays each one for two seconds; the green dot is the turning point.

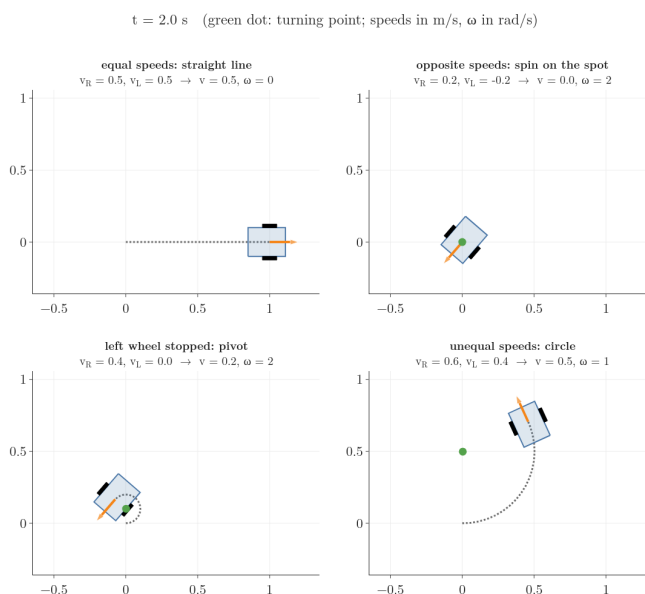


Figure 5: Four motions of the robot, each played for two seconds with its trail: equal wheel speeds drive straight; opposite speeds spin in place; a stopped left wheel pivots about that wheel; unequal speeds drive on a circle of radius 0.5 m

Wheel speeds				
v_R, v_L (m/s)	v (m/s)	ω (rad/s)	R (m)	Motion
0.5, 0.5	0.5	0	none	straight line
0.2, -0.2	0	2	0	spin on the spot
0.4, 0	0.2	2	0.1	pivot about the left wheel
0.6, 0.4	0.5	1	0.5	circle to the left

Why each happens:

- **Straight line.** Equal speeds have no difference, so $\omega = 0$: nothing turns the heading. The turning point is infinitely far away.

- **Spin on the spot.** Opposite speeds add up to zero, so $v = 0$: the midpoint stays put while the heading turns.
- **Pivot.** A stopped wheel does not move, so it must be the turning point itself; the table confirms a turning radius of 0.1 m, exactly half the wheel separation.

The spin on the spot is what makes the differential drive so easy to steer: it can face any direction first and then drive straight (LaValle §13.1.2.2). A car cannot (Section 7).

4.5 Back again: the wheel speeds for a wanted motion

Key point: To drive with forward speed v and turn rate ω , run the right wheel at $v + \omega L/2$ and the left wheel at $v - \omega L/2$.

The control question of Section 1 goes the other way: the controller has chosen v and ω and must send wheel commands. This is the **inverse kinematics** (G-2295) of the robot. Solve the two formulas of Section 4.3 for the wheel speeds:

$$v_R = v + \omega L/2$$

$$v_L = v - \omega L/2$$

Each wheel runs at the midpoint's speed, plus or minus the extra speed it needs for being half an axle farther out or closer in. Then divide by the wheel radius to get the spin rates for the motors.

Worked example. We want 0.3 m/s while bearing left at 0.5 rad/s. The extra speed for half an axle:

$$\omega L/2 = 0.5 \times 0.1 = 0.05 \text{ m/s}$$

The wheel speeds:

$$v_R = 0.3 + 0.05 = 0.35 \text{ m/s}$$

$$v_L = 0.3 - 0.05 = 0.25 \text{ m/s}$$

The spin rates:

$$u_R = 0.35/0.05 = 7 \text{ rad/s}$$

$$u_L = 0.25/0.05 = 5 \text{ rad/s}$$

As a check, feeding 7 and 5 rad/s back into the forward kinematics of Section 4.3 must return 0.3 m/s and 0.5 rad/s, and it does (the Python box below).

Python: both directions in four lines.

```
r, L = 0.05, 0.2          # wheel radius, wheel separation (m)
def forward(uR, uL):       # spin rates (rad/s) -> v (m/s), omega
    ↪ (rad/s)
    return r * (uR + uL) / 2, r * (uR - uL) / L
def inverse(v, omega):     # v, omega -> spin rates (rad/s)
    return (v + omega * L / 2) / r, (v - omega * L / 2) / r
inverse(0.3, 0.5)         # (7.0, 5.0)
forward(7, 5)              # (0.3, 0.5): back where we started
```

5. From robot speeds to motion on the floor

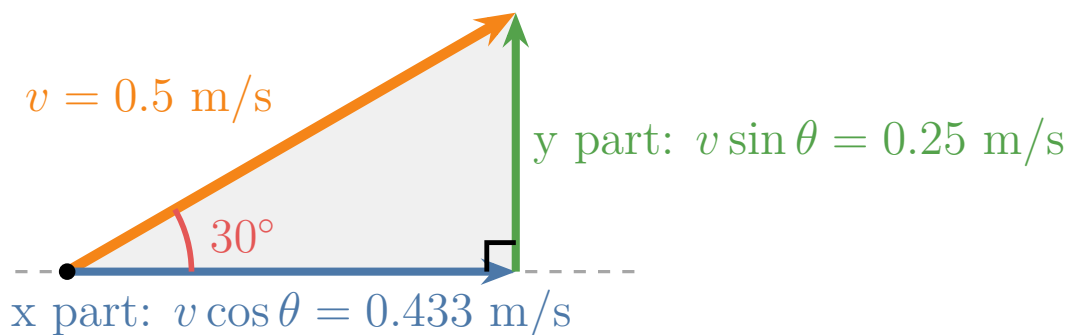
Key point: The robot moves along its heading, so its forward speed splits into an x part $v \cos \theta$ and a y part $v \sin \theta$. Because the heading keeps changing, we predict the path in many small steps.

5.1 Why the speed splits into cosine and sine parts

The robot knows its forward speed, but the map needs to know how fast x and y change. The robot moves in the direction it faces, so the share of its speed that goes into x and into y depends on the heading. The two extremes are easy:

- facing along x ($\theta = 0^\circ$): all of the speed goes into x , none into y ;
- facing along y ($\theta = 90^\circ$): all of it goes into y , none into x .

In between, the speed is shared. Draw the velocity as an arrow of length v at angle θ (Figure 6). It is the long side of a right-angled triangle whose two short sides are the x speed and the y speed.



robot at the origin of the arrow, facing $\theta = 30^\circ$

Figure 6: The velocity arrow of length 0.5 m/s at 30 degrees, with its x part 0.433 m/s and its y part 0.25 m/s drawn as the two short sides of a right-angled triangle

The cosine and sine of an angle are defined as exactly these shares of the long side:

- $\cos \theta$ is the side along x divided by the long side;
- $\sin \theta$ is the side along y divided by the long side.

So the x speed is $v \cos \theta$ and the y speed is $v \sin \theta$. The extremes check out: $\cos 0^\circ = 1$ and $\sin 0^\circ = 0$ (everything along x); $\cos 90^\circ = 0$ and $\sin 90^\circ = 1$ (everything along y).

Worked example. At $\theta = 30^\circ$, $\cos 30^\circ = 0.866$ and $\sin 30^\circ = 0.5$. With $v = 0.5$ m/s:

$$\text{x speed} = v \cos \theta$$

$$= 0.5 \times 0.866 = 0.433 \text{ m/s}$$

$$\text{y speed} = v \sin \theta$$

$$= 0.5 \times 0.5 = 0.25 \text{ m/s}$$

The two short sides must rebuild the long side (Pythagoras), and they do:

$$0.433^2 + 0.25^2 = 0.25$$

$$\sqrt{0.25} = 0.5 \text{ m/s}$$

5.2 The kinematic model

Key point: Three equations give how fast each number of the pose changes, from the two inputs v and ω .

We need a short way to write “how fast this changes”. A dot over a symbol means its rate of change per second, the **derivative** with respect to time (G-2296). For example, $\dot{x} = 0.433$ m/s means x grows by 0.433 m each second.

The x speed and y speed of Section 5.1 are $\dot{x}$ and $\dot{y}$. The heading changes at the turn rate, by the definition of ω in Section 4.1. Together:

$$\dot{x} = v \cos \theta$$

$$\dot{y} = v \sin \theta$$

$$\dot{\theta} = \omega$$

This is the **kinematic model** (G-2297) of the differential-drive robot. “Kinematic” means it uses speeds only, not masses or forces. That is safe while the robot moves slowly, because then the wheels follow their commands closely; at high speed, skidding and inertia matter and a dynamic model is needed (LaValle §13.1.2.1). The two numbers we choose, $u = (v, \omega)$, are the **control inputs** (G-2298). A sanity check: at $\theta = 0$ the model gives $\dot{x} = v$ and $\dot{y} = 0$, straight along x, as it should.

The same three equations describe a single rolling wheel, the unicycle (MR §13.3.1); a later Note in this chapter compares wheel types.

The three lines can be written as one matrix times the input vector:

$$\dot{q} = G(q) u$$

$$\dot{q} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$$

$$G(q) = \begin{bmatrix} \cos \theta & 0 \\ \sin \theta & 0 \\ 0 & 1 \end{bmatrix}$$

$$u = \begin{bmatrix} v \\ \omega \end{bmatrix}$$

The matrix form shows something the three lines hide. Each column of $G(q)$ is one direction the robot can move in: the first column is “drive forward”, the second is “turn on the spot” (MR §13.3.1). Every motion is a mix of these two. A third direction, sliding sideways, is missing, and Section 6 is about that gap.

At our pose, with the input $u = (0.5, 1)$, the matrix product gives back the speeds of Section 5.1:

$$\dot{x} = 0.866 \times 0.5 + 0 \times 1 = 0.433$$

$$\dot{y} = 0.5 \times 0.5 + 0 \times 1 = 0.25$$

$$\dot{\theta} = 0 \times 0.5 + 1 \times 1 = 1$$

5.3 Why we predict the path step by step

Key point: The direction of motion changes as the heading turns, so we cannot just multiply speed by time. Over a short step the heading hardly changes, so we treat it as fixed, move, and repeat.

Speed times time gives distance only if the direction stays the same. Here the heading turns at 1 rad/s, so the direction of motion changes all the time. But over a short time step Δt the heading changes only a little, so we can treat the rates as fixed during the step and add them up. This is an **Euler step** (G-2299):

$$x_{\text{new}} = x + \dot{x} \Delta t$$

$$y_{\text{new}} = y + \dot{y} \Delta t$$

$$\theta_{\text{new}} = \theta + \dot{\theta} \Delta t$$

Worked example. Start at the pose of Figure 1, keep $v = 0.5$ m/s and $\omega = 1$ rad/s, and take steps of $\Delta t = 0.1$ s. The first step:

$$x = 2 + 0.433 \times 0.1 = 2.0433$$

$$y = 1 + 0.25 \times 0.1 = 1.0250$$

$$\theta = 0.5236 + 1 \times 0.1 = 0.6236$$

The heading has changed, so the next step uses new rates:

Step	x (m)	y (m)	θ (rad)	$\dot{x}$ (m/s)	$\dot{y}$ (m/s)
0	2.0000	1.0000	0.5236	0.4330	0.2500
1	2.0433	1.0250	0.6236	0.4059	0.2920
2	2.0839	1.0542	0.7236	0.3747	0.3310

Figure 7 keeps going. The path bends left, and after about 6.3 s (63 steps) it closes a circle of radius 0.5 m, the turning radius of Section 4.3.

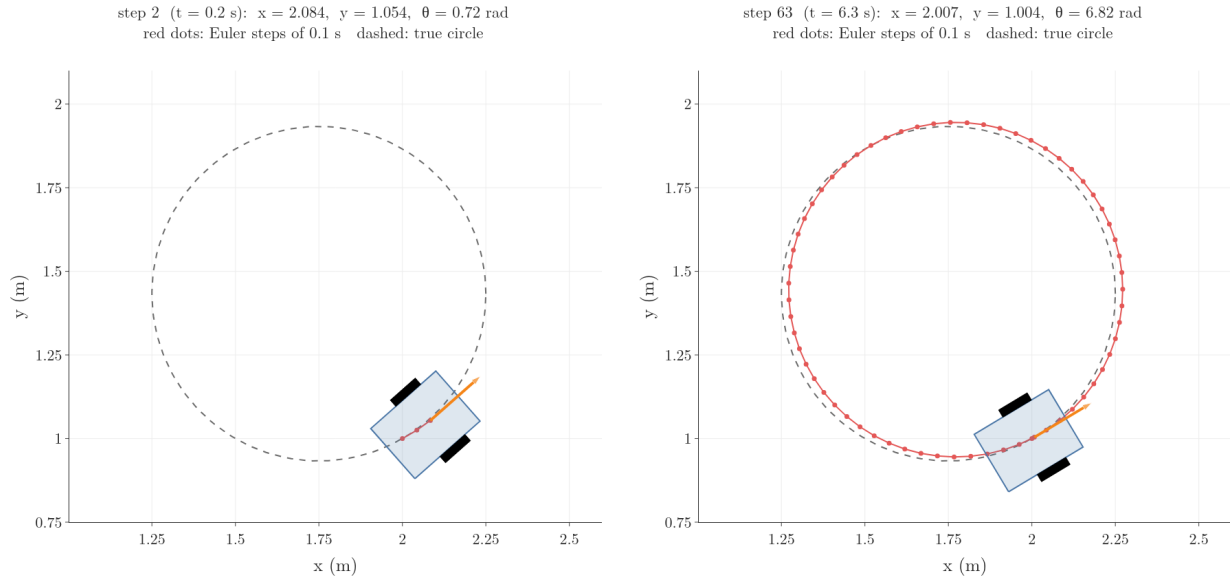


Figure 7: The robot driving from pose (2, 1, 30 degrees) with $v = 0.5$ m/s and $\omega = 1$ rad/s; dots mark every Euler step of 0.1 s and the trail closes a circle of radius 0.5 m

The prediction is not perfect. Each step moves along a straight line in the direction the robot faced at the start of the step, while the true robot curves. So the predicted path drifts from the true circle (the dashed line in Figure 7). The smaller the step, the less the heading changes within it, and the smaller the drift. The largest gap between the predicted and the true position during one lap:

Step Δt	Largest gap
1 s	51 cm
0.5 s	25 cm
0.1 s	5 cm
0.01 s	0.5 cm

The gap shrinks in step with Δt : a step ten times smaller gives a gap ten times smaller. The

notebook `R0-001-pose-and-wheeled-robot-motion.ipynb` reproduces this table and lets us try other wheel speeds.

6. The robot cannot slide sideways

Key point: Rolling wheels forbid sideways motion. That limits which way the robot can move at each moment, but not which poses it can reach; it only means some moves need manoeuvres.

Why does this deserve a section? Many route planners assume a robot can move in any direction from where it stands. A route that needs a step sideways cannot be followed by our robot or by a car, so those planners cannot be used as they are (MR §13.3.3). To plan for wheeled robots, we must state the rule exactly and understand what it does and does not forbid.

6.1 The rule as an equation

A wheel rolls along its own direction but grips against sliding sideways (LaValle §13.1.2.1). So the robot's midpoint can move forward or backward, never along its own left-right axis. In words: **the robot's velocity has no part pointing to its left.**

To turn that sentence into an equation, we need the robot's left direction in world coordinates. Its forward direction at heading θ is the arrow $(\cos \theta, \sin \theta)$, by Section 5.1 with a speed of 1. Its left is the forward direction turned by 90° , at heading $\theta + 90^\circ$ (Figure 8):

$$(\cos(\theta + 90^\circ), \sin(\theta + 90^\circ))$$

$$= (-\sin \theta, \cos \theta)$$

At $\theta = 0$ this gives $(0, 1)$, the y-axis: the left of a robot facing along x.

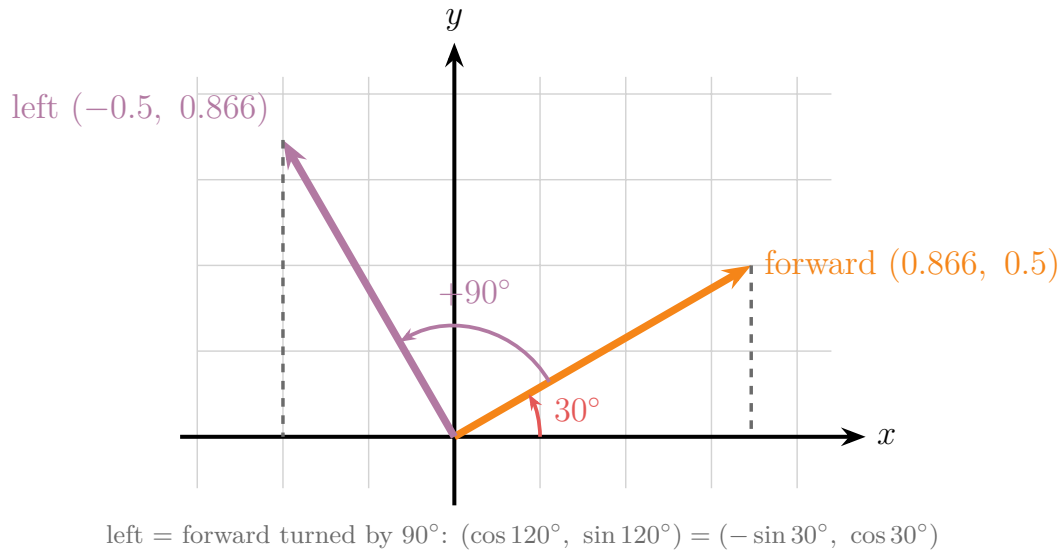


Figure 8: The forward arrow $(0.866, 0.5)$ of a robot at 30° and its left arrow $(-0.5, 0.866)$, the forward arrow turned by 90°

How much of a velocity points along an arrow of length 1 is found by multiplying matching entries and adding, the **dot product** (G-634). For the velocity $(\dot{x}, \dot{y})$ and the left arrow, the rule says this must be zero:

$$-\dot{x} \sin \theta + \dot{y} \cos \theta = 0$$

Check an allowed motion. Driving forward at our pose $(\dot{x} = 0.433, \dot{y} = 0.25, \theta = 30^\circ)$:

$$-0.433 \times 0.5 = -0.2165$$

$$0.25 \times 0.866 = 0.2165$$

$$-0.2165 + 0.2165 = 0$$

No part points left: allowed.

Check a forbidden motion. Moving straight to the robot's left at 0.5 m/s $(\dot{x} = -0.25, \dot{y} = 0.433)$:

$$-(-0.25) \times 0.5 = 0.125$$

$$0.433 \times 0.866 = 0.375$$

$$0.125 + 0.375 = 0.5$$

All 0.5 m/s points left: the wheels would have to skid.

The rule can be written as a row of numbers times $\dot{q}$:

$$A(q) \dot{q} = 0$$

$$A(q) = \begin{bmatrix} -\sin \theta & \cos \theta & 0 \end{bmatrix}$$

A **constraint** (a condition that must always hold) on velocities, written in this form $A(q) \dot{q} = 0$, is a **Pfaffian constraint** (G-2300) (MR §2.4). The form is worth a name because rules from very different machines fit it, such as a rolling coin, a car and a robot hand holding an object, and one set of tools then handles all of them (MR §2.4).

The constraint and the kinematic model say the same thing from two sides. The model lists the directions the robot can move in, the columns of $G(q)$; the constraint names the one it cannot. So every column of $G(q)$ must pass the constraint. At $\theta = 30^\circ$, the first column (0.866, 0.5, 0):

$$-0.5 \times 0.866 + 0.866 \times 0.5 + 0 = 0$$

The second column (0, 0, 1):

$$-0.5 \times 0 + 0.866 \times 0 + 0 \times 1 = 0$$

6.2 Why the robot can still reach every pose: holonomic and nonholonomic constraints

Key point: A holonomic constraint fixes where a system can be. A nonholonomic constraint only fixes which way it can move at each moment; the system can still reach every configuration, with manoeuvres.

Does “no sideways motion” mean there are places the robot can never get to? Compare it with a system where that really is so: a toy train on a circular track of radius 5 m (Figure 9, left). The train’s position obeys a rule about **position**:

$$x^2 + y^2 = 25$$

At the point (3, 4), the check is:

$$3^2 + 4^2 = 9 + 16 = 25$$

The train can never leave the circle. The position rule also limits its velocity: as long as $x^2 + y^2$ stays at 25, its rate of change must be zero. By the [chain rule](#) (G-371), the rate of change of x^2 is $2x$ times the rate of change of x . So:

$$2x \dot{x} + 2y \dot{y} = 0$$

At (3, 4):

$$6 \dot{x} + 8 \dot{y} = 0$$

This velocity rule has the same Pfaffian form as the robot's, but it comes from a position rule. A rule on position, or a velocity rule that comes from one, is a **holonomic constraint** (G-2301). It removes whole configurations: the train lives on a 1-dimensional circle, not on the 2-dimensional floor (MR §2.4).

holonomic: train on a track nonholonomic: rolling robot

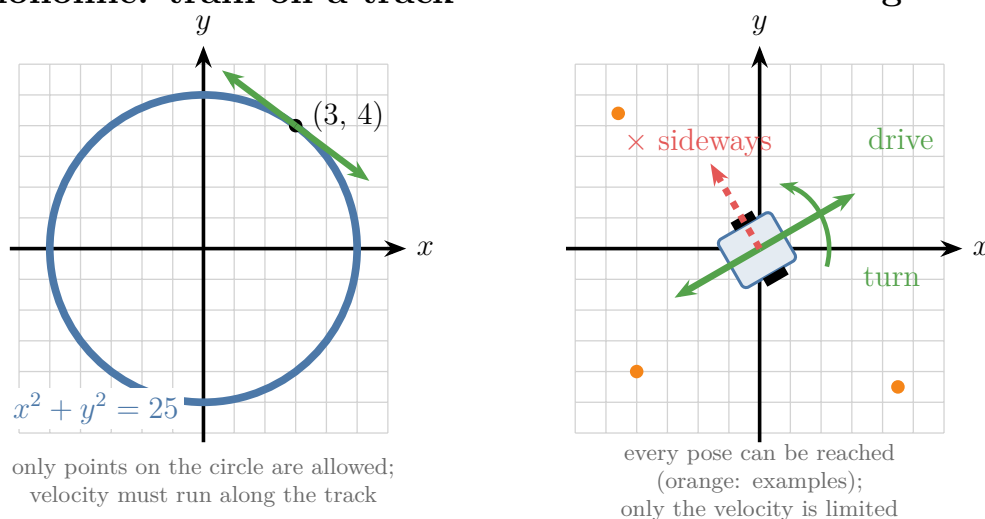


Figure 9: Left: a train on a circular track of radius 5 m can only be on the circle. Right: the robot at one pose can only move along its heading, yet it can reach any pose on the floor

Is there a position rule hiding behind the robot's constraint? Suppose there were one, linking x , y and θ . Then fixing the position (x, y) would also fix the heading θ , the way fixing x on the track fixes y . But the robot can spin on the spot (Section 4.4) and point in any direction from the same position. So no such rule exists (MR §2.4).

Figure 10 shows the consequence. Three legal moves shift the robot 0.5 m to its left, the one direction it can never move in directly:

1. Spin 90° to the left on the spot.
2. Drive 0.5 m forward.

3. Spin 90° back to the right.

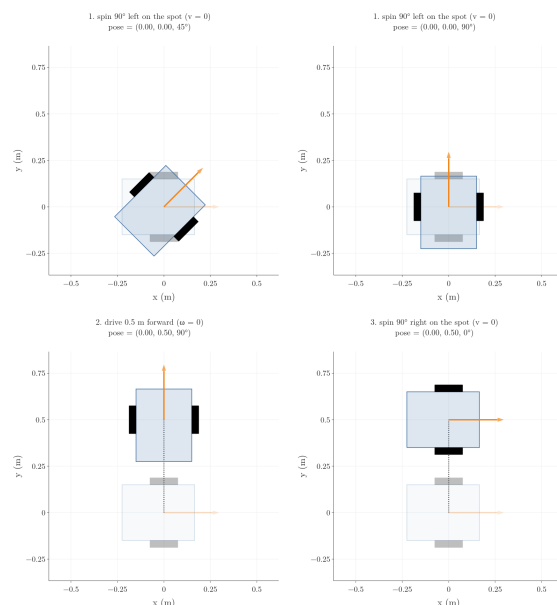


Figure 10: The robot moves 0.5 m sideways using only legal moves: spin 90 degrees left, drive 0.5 m, spin 90 degrees right. It ends 0.5 m to the left of where it started, facing the same way

A velocity rule that does not come from any position rule is a **nonholonomic constraint** (G-2302). It cuts the directions the robot can move in at each moment from 3 to 2, but the robot can still reach every pose (MR §2.4). Robots with such constraints are called nonholonomic robots. A car shows the same thing every day: it cannot drive sideways into a parking space, but it gets there by parallel parking (LaValle §13.1.2.1).

	Holonomic constraint	Nonholonomic constraint
Limits	positions (and so velocities)	velocities only
Example	train on a circular track	wheel that cannot slide sideways
Reachable configurations	fewer: the track only	all of them
Rule	$g(q) = 0$, such as $x^2 + y^2 = 25$	$A(q) \dot{q} = 0$ with no $g(q)$ behind it
What it means for planning	search a smaller space	every pose is reachable, but routes need manoeuvres

Extra: Modern Robotics states the test in algebra. A Pfaffian constraint $A(q) \dot{q} = 0$ is holonomic exactly when $A(q)$ is the matrix of partial derivatives (the **Jacobian**) of some function $g(q)$, so that it “integrates” back to $g(q) = 0$. For the train, $A = [2x, 2y]$ is the Jacobian of the track rule $g(x, y)$, where:

$$g(x, y) = x^2 + y^2 - 25$$

For the rolling robot, no such g exists, so the constraint is called nonintegrable (MR §2.4).

7. Cars: the same model with different limits

Key point: A differential-drive robot and a car follow the same three equations. They differ only in which pairs of forward speed and turn rate they can produce, and that one difference explains why a car cannot turn on the spot.

7.1 Why one model covers cars too

A car's rear wheels also cannot slide sideways, and its steered front wheels make it turn about a point on the line through its rear axle, for the same reason as in Section 4.2. So the midpoint of its rear axle follows the same model as our robot (MR §13.3.1):

$$\dot{x} = v \cos \theta$$

$$\dot{y} = v \sin \theta$$

$$\dot{\theta} = \omega$$

The difference lies elsewhere: not every pair (v, ω) is possible for every vehicle. All the pairs a vehicle can produce form its **control set** (G-2309). Figure 11 draws each control set in the plane of forward speed (across) and turn rate (up); the shape of the set decides what the vehicle can do.

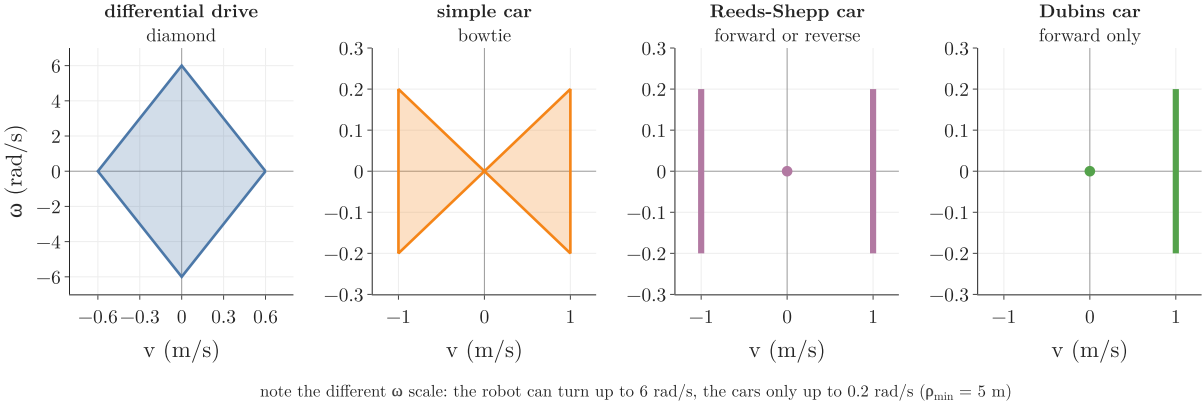


Figure 11: Four control sets in the plane of forward speed v and turn rate ω : a diamond for the differential drive, a bowtie for the simple car, two vertical segments for the Reeds-Shepp car and one segment for the Dubins car

Differential drive: why a diamond. Each wheel has a top speed, say 0.6 m/s. By Section 4.5, each wheel speed is a straight-line mix of v and ω , so each limit cuts the plane along a straight line, and the four limits (each wheel, forward and backward) make a four-sided shape. Its corners are the extremes. Driving straight with both wheels at 0.6 m/s gives the top forward speed:

$$v = 0.6 \text{ m/s}$$

Spinning with the wheels at +0.6 and −0.6 m/s gives the top turn rate:

$$\begin{aligned} \omega &= \frac{0.6 - (-0.6)}{0.2} \\ &= 6 \text{ rad/s} \end{aligned}$$

The corner at no forward speed and a turn rate of 6 rad/s is the spin on the spot.

Simple car: why a bowtie. The **simple car** (G-2303) steers its front wheels, but only so far, so it cannot drive on a circle tighter than its **minimum turning radius** (G-2304) $\rho_{\min}$ (LaValle §13.1.2.1). A tighter circle means a faster turn at the same speed, because $R = v/\omega$. So the turn rate is limited by the speed:

$$|\omega| \leq |v|/\rho_{\min}$$

With $\rho_{\min} = 5 \text{ m}$ and a top speed of 1 m/s, the largest turn rate is:

$$\omega_{\max} = 1/5 = 0.2 \text{ rad/s}$$

The limit is two straight lines through the origin, which gives the bowtie. At zero speed the allowed turn rate is zero: a car turns only by rolling along a curve, so a car standing still cannot turn, and it can never spin on the spot. Where $\rho_{\min}$ comes from (the steering angle and the distance between the axles) is the subject of the car-like robots Note later in this chapter.

Reeds-Shepp car and Dubins car: why simplify further. The **Reeds-Shepp car** (G-2305) is a simple car that drives only at full speed forward or full speed in reverse ($v = +1$ or $v = -1$), or stands still, like the gears forward, reverse and park. The **Dubins car** (G-2306) is the same without reverse: $v = +1$ or stop (LaValle §13.1.2.1). Why study such restricted cars?

- Limiting the speed to full forward or full reverse does not remove any pose the car can reach; it only fixes how fast it gets there (LaValle §13.1.2.1).
- For these two cars, the shortest route between any two poses on an open floor is known exactly: it is always made of arcs at the minimum turning radius and straight lines, from a short list of patterns (MR §13.3.3). Planners use these routes as ready-made building blocks; a later Note on planning with motion limits does so.

7.2 What the limits change

All four vehicles are nonholonomic and can reach every pose on an open floor (LaValle §13.1.2.1). The control set decides how hard it is to get there:

- The differential drive can spin on the spot, so it can turn to face the goal first and then drive straight to it (LaValle §13.1.2.2).
- The Reeds-Shepp car can get into an arbitrarily small parking space, given a little clearance, by shuffling forward and back (LaValle §13.1.2.1).
- The Dubins car cannot reverse, so parallel parking in a tight space is impossible. Facing a wall, it may be unable to avoid hitting it (LaValle §13.1.2.1).

8. A car pulling a trailer

Key point: Each trailer adds one number to the configuration, its own heading, but no new control input. Pulled forward, the trailer swings into line behind the car; pushed in reverse, the angle between them grows.

8.1 Why the trailer turns

Hitch a trailer to the middle of the car's rear axle (Figure 12). The car decides where the hitch goes; the trailer can only follow. To know where everything is, we need one more number, the trailer's heading θ_1 ; we now call the car's heading θ_0 . The configuration has four numbers:

$$q = (x, y, \theta_0, \theta_1)$$

The distance from the hitch to the middle of the trailer's axle is the **hitch length** (G-2307) d_1 .

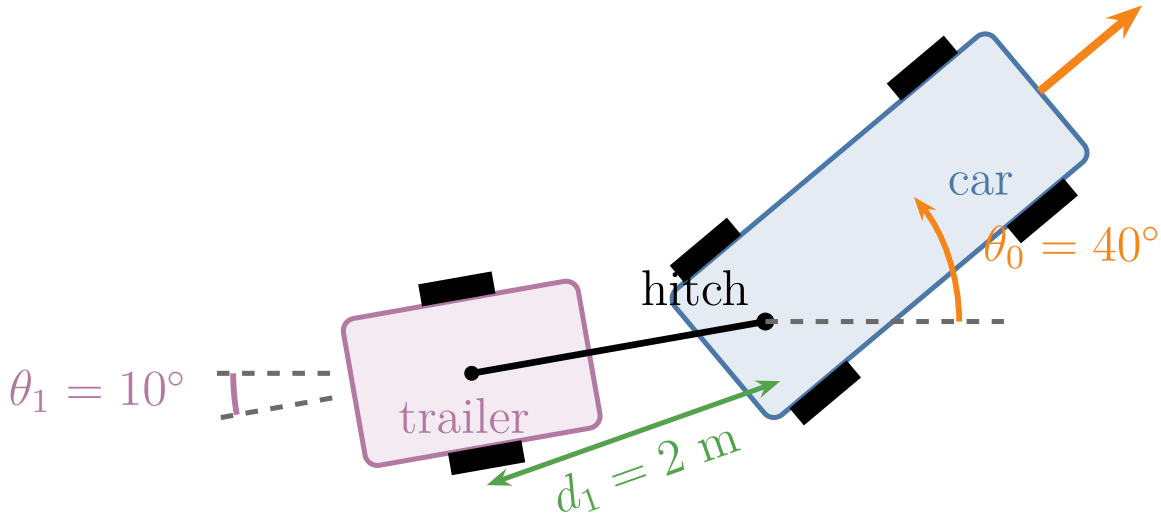
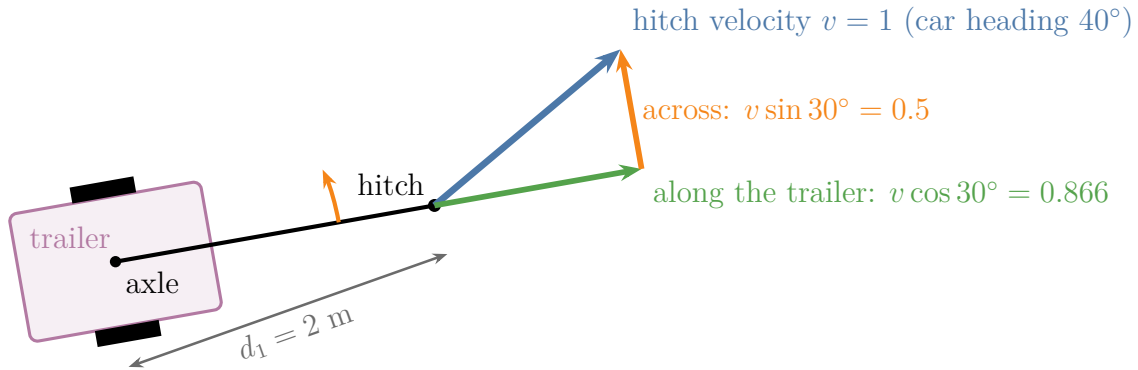


Figure 12: A car seen from above pulling one trailer: the car heading of 40 degrees, the trailer heading of 10 degrees, the hitch point at the middle of the car's rear axle and the hitch length of 2 m

How fast does the trailer's heading change? Take the car heading at 40° , the trailer at 10° , $v = 1$ m/s and $d_1 = 2$ m. The hitch moves with the car: 1 m/s in the car's direction. Split that velocity into two parts relative to the trailer (Figure 13), using the angle between car and trailer:

$$\theta_0 - \theta_1 = 40^\circ - 10^\circ = 30^\circ$$

- the part **along** the trailer, $v \cos 30^\circ = 0.866$ m/s, pulls the trailer forward;
- the part **across** the trailer, $v \sin 30^\circ = 0.5$ m/s, pushes the front of the trailer sideways.



the axle cannot slide sideways, so the across part swings the trailer about its axle:
 turn rate = across speed $\div$ distance = $0.5 \div 2 = 0.25 \text{ rad/s}$

Figure 13: The hitch velocity of 1 m/s along the car's heading split into a part along the trailer, 0.866 m/s, and a part across it, 0.5 m/s; the across part swings the trailer about its axle

The trailer's own wheels cannot slide sideways, so its axle does not move across. The front end moves across at 0.5 m/s while the axle, 2 m behind, does not: the trailer turns about its axle. By the rule of Section 4.2, speed equals turn rate times distance, so the turn rate is the across speed divided by the distance:

$$\dot{\theta}_1 = \frac{0.5}{2} = 0.25 \text{ rad/s}$$

In general:

$$\dot{\theta}_1 = \frac{v}{d_1} \sin(\theta_0 - \theta_1)$$

This is the trailer equation of LaValle §13.1.2.4 (eq. 13.19). The car's own three equations stay as in Section 7.1.

The formula explains what drivers see. The trailer's heading grows towards the car's heading, so the gap shrinks. As the gap shrinks, the across part shrinks with it, and when the two are in line the across part is zero and the trailer stops turning. A longer hitch swings more slowly, because the same across speed has a longer lever to turn.

8.2 Why reversing is hard: jackknifing

Now reverse, with $v = -1 \text{ m/s}$ and the same angles:

$$\dot{\theta}_1 = \frac{-1}{2} \times 0.5 = -0.25 \text{ rad/s}$$

The sign flips: the trailer now turns **away** from the car's heading, so the gap grows. A bigger gap makes the across part bigger, which makes the trailer turn away faster still. Left alone, the car and trailer fold into a V, called **jackknifing** (G-2308). Figure 14 shows both cases from the same start: forward, the gap falls from 30° to 2° in 6 s; in reverse, it grows from 30° to 89° in 2.6 s.

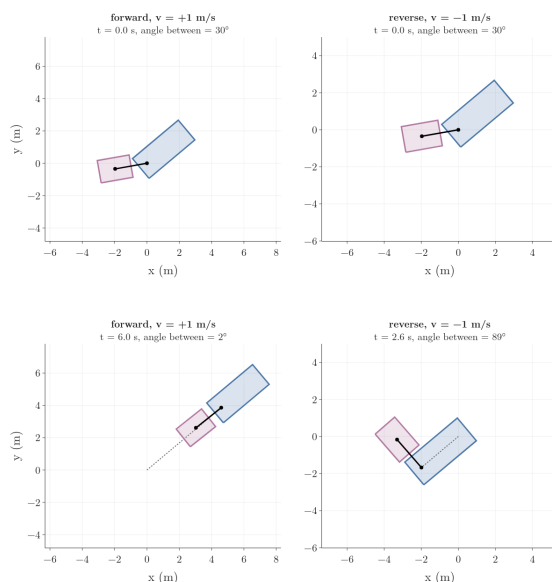


Figure 14: The car and trailer start with a 30 degree angle between them. Driving forward, the trailer lines up behind the car. Reversing, the angle grows and the pair folds up

A rope behaves the same way. Pull a rope and it trails in line behind you; push it and it buckles. A trailer in reverse is being pushed, so the driver has to steer constantly to keep it from folding. With k trailers the configuration has $3 + k$ numbers, but there are still only two inputs, v and ω (LaValle §13.1.2.4), so each extra trailer is one more angle to keep in check with the same two controls.

9. Summary

Robot	Configuration	Inputs	Control set	So it can
Differential drive	(x, y, θ)	v, ω from two wheel speeds	diamond	spin on the spot, then drive straight
Simple car	(x, y, θ)	v, ω	bowtie: $ \omega \leq v /\rho_{\min}$	turn only while moving
Reeds-Shepp car	(x, y, θ)	$v = \pm 1$ or $0, \omega$	two segments	park by shuffling forward and back
Dubins car	(x, y, θ)	$v = 1$ or $0, \omega$	one segment	reach any pose on an open floor, but not park tightly
Car with one trailer	$(x, y, \theta_0, \theta_1)$	v, ω	as the car	pull the trailer into line; reversing folds it

- The pose (x, y, θ) says where a floor robot is, because position alone does not say which way it will drive off.
- Angles are in radians, so a turning point's arc is radius times angle, with no conversion factor.
- A wheel that rolls without slipping moves at r times its spin rate, because it lays its rim down on the floor.
- The forward speed is the average of the wheel speeds and the turn rate is their difference over L , because the rigid robot turns about one point on its axle line and each point's speed is turn rate times distance.
- Controllers think in v and ω , and the inverse formulas turn those into wheel commands, so the controller never has to reason in wheel speeds.
- The kinematic model gives the rates of x, y and θ ; the heading keeps turning, so we predict the path in small Euler steps, and smaller steps drift less.
- The wheels forbid sideways motion, a nonholonomic constraint: it removes one direction of motion at each moment but no reachable pose, so the robot gets anywhere with manoeuvres such as spin–drive–spin or parallel parking.
- Cars follow the same equations with a smaller control set, so they need more manoeuvring: they turn only while moving, and the Dubins car cannot even reverse.
- A trailer adds a heading but no input; pulling swings it into line and pushing folds it, so reversing needs constant correction.

So the two questions of the opening have their answers: the forward formulas and Euler steps predict where wheel commands take the robot, and the inverse formulas give the commands for a chosen motion.

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11. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Coordinate frame (G-2281)	A chosen origin and set of axes used to give positions as numbers; every position or pose is stated relative to some frame.
World frame (G-2282)	The coordinate frame fixed to the surroundings (for example a corner of the room), in which the robot’s pose is given.
Heading θ (G-2284)	The angle from the world x-axis to the direction a robot faces, measured counter-clockwise; it decides which way the robot moves when it drives forward.

Term	Meaning
Pose (G-2280)	Where a robot is and which way it faces: on a floor, its position (x, y) and heading θ , such as (2 m, 1 m, 30°); needed because a robot at one spot drives off differently depending on its heading.
Radian (G-2285)	The unit of angle at which the arc of a circle is as long as its radius; a full turn is 2π rad, and with radians an arc is simply radius times angle.
Body frame (G-2283)	A coordinate frame attached to the robot, with its x-axis pointing forward and its y-axis to the left; it moves and turns with the robot, so things can be described from the robot's point of view.
Configuration (G-2286)	The list of numbers that pins down where every part of a robot is, such as $q = (x, y, \theta)$ for a floor robot; planning and models work with it as one vector.
Degrees of freedom (G-2310)	The number of values in a robot's configuration, such as 3 for a floor robot (x, y, θ) : how many independent ways it can be placed.
Configuration space (C-space) (G-2287)	The set of all possible configurations of a robot; for a floor robot, any (x, y) with a heading on a circle.
Differential drive (G-2288)	A robot with two independently driven wheels on one axle (plus a caster); it steers by running the wheels at different speeds and can spin on the spot.
Caster wheel (G-2289)	A free-swivelling, undriven wheel, like the one under an office chair; it supports a robot without steering it.
Rolling without slipping (G-2290)	A wheel moving so that its contact point does not skid: it moves forward by its radius times the angle it turns and never slides sideways, which is what lets wheel turns predict motion.
Forward speed v (G-2291)	How fast a robot's reference point (for a differential drive, the middle of the axle) moves along its heading, in m/s; for a differential drive, the average of the two wheel speeds.

Term	Meaning
Turn rate ω (G-2292)	How fast a robot's heading changes, in rad/s, positive to the left; for a differential drive, the difference of the wheel speeds divided by the wheel separation.
Turning radius R (G-2293)	The distance from the point a robot is turning about (the turning point, or instantaneous centre of curvature) to the robot's reference point; $R = v/\omega$, so it says how tight a turn is.
Forward kinematics (wheeled robot) (G-2294)	Working out the robot's forward speed and turn rate from its wheel speeds, used to predict where the robot goes.
Inverse kinematics (wheeled robot) (G-2295)	Working out the wheel speeds that give a wanted forward speed and turn rate, used by a controller to command the motors.
Dot notation $\dot{x}$ (G-2296)	A dot over a quantity means its rate of change per second (its derivative with respect to time), such as $\dot{x} = 0.433$ m/s; models of motion are written with it.
Kinematic model (G-2297)	Equations that give how a robot's configuration changes from its speed inputs alone, ignoring masses and forces, such as $\dot{x} = v \cos \theta$, $\dot{y} = v \sin \theta$, $\dot{\theta} = \omega$; used to predict motion at low speed.
Control input u (G-2298)	The numbers we choose to drive a robot, such as $u = (v, \omega)$; the model turns them into motion.
Euler step (G-2299)	One step of predicting motion: hold the rates fixed for a short time Δt and add rate times Δt to each quantity; repeated, it traces a path, more accurately the smaller Δt is.
Pfaffian constraint (G-2300)	A rule on a system's velocities of the form $A(q) \dot{q} = 0$, such as the robot's no-sideways rule $-\dot{x} \sin \theta + \dot{y} \cos \theta = 0$; it marks which directions of motion are forbidden.
Holonomic constraint (G-2301)	A rule on positions, $g(q) = 0$, such as a train on a circular track $x^2 + y^2 = 25$ (or a velocity rule that comes from one); it removes whole configurations the system can never be in.

Term	Meaning
Nonholonomic constraint (G-2302)	A velocity rule that does not come from any position rule, such as a wheel that cannot slide sideways; it limits which way a robot can move at each moment but not which configurations it can reach.
Control set (G-2309)	All the input pairs (v, ω) a vehicle can produce, drawn as a shape (diamond, bowtie, segments); the shape decides what manoeuvres are possible, such as spinning on the spot.
Simple car (G-2303)	A car model that moves like the differential drive but cannot turn on a circle smaller than its minimum turning radius, so its turn rate is limited by its speed and it cannot spin on the spot.
Minimum turning radius $\rho_{\min}$ (G-2304)	The radius of the tightest circle a car can drive, set by how far it can steer; it limits the turn rate to $ \omega \leq v /\rho_{\min}$.
Reeds-Shepp car (G-2305)	A simple car that moves at full speed forward or full speed in reverse (or stands still); with reverse it can manoeuvre into tight spaces.
Dubins car (G-2306)	A simple car that can only move forward at full speed (or stop); it can reach any pose on an open floor but cannot parallel park in a tight space.
Hitch length d_1 (G-2307)	The distance from the hitch point to the middle of a trailer's axle; a longer hitch makes the trailer's heading change more slowly.
Jackknifing (G-2308)	A car and trailer folding into a V when reversing, because in reverse the angle between them grows instead of shrinking.
Dot product (G-634)	Multiplying two vectors entry by entry and adding the products, $u^T x$, which gives one number. It is large when the vectors point the same way, so it measures similarity and gives projections.
Chain rule (G-371)	To differentiate a function of a function, multiply the outer derivative by the inner derivative.